Jerry Xizhao Fu

Montreal, QC, Canada | French, English, Mandarin
jerryfu_mc@outlook.com | +1 (438) 225-8853 | jerryxf.net

Student building intelligent systems across robotics, medicine, and AI.

Education

2025–(2027) Collège Jean-de-Brébeuf – IBDP + DEC Pure and Applied Sciences (pre-uni)
2020–2025 Collège Regina Assumpta – Secondary I-V, Natural Sciences and Robotics

Research Experience

2025-present CHUM Research Center – Research Assistant and Clinical Observership

- Analyzed regional pleural strain via lung ultrasound elastography to establish normative values and their correlations with tidal volume in healthy volunteers.
- Shadowed in the Intensive Care Unit (ICU) and Operating Room; exposure to anesthesia, critical care, and surgical workflows.

Publications/Abstracts

CHUM Research Center - SFAR Congress

- *Relation entre la déformation pleurale régionale mesurée par élastographie pulmonaire et le volume courant chez des volontaires sains.* S. Chraïbi, K. El Berhoumi, A. Danatoiu, **J. Fu**, P. Vianna, M.H. Roy Cardinal, G. Cloutier, M. Girard (presenting author). Presented at the 2025 Annual Congress of the French Society of Anesthesia and Critical Care (SFAR) on Sep 17, 2025, in Paris, France.

Work Experience

2026 Unveil Technologies – Founder & Lead Developer

- Designed and built ORCA, an operational intelligence platform that fuses multi-source data into a live, explainable operational picture for decision support — https://unveiltechnologies.com
- Architected a distributed pipeline ingesting telemetry, sensor, and radio data through NATS, a streaming entity graph, and vector search — producing real-time, cited intelligence assessments.
- Led design, branding, and a TRL-3 funding proposal for the Canadian DND/CAF IDEaS program.

2022–present FIRST Robotics Competition – Mentor & former Student of Team 3990

- Created coursework and lectured programming, computer vision and AI/ML to junior team members.
- Prepared a whitepaper on computer vision in robotics.
- Developed computer vision object tracking and autonomous behaviors on the robot.
- Multiple consecutive Regional Winners; 2024 World Championship Impact Award finalist.

2023-2024 DODO Wheels – Sole Software Developer

- 2024 summer | Developed Doublestar Tire's Canadian website — https://doublestartyre.ca
- 2023 summer | Managed, updated, and populated internal inventory through CPanel.

Volunteer Experience

- 2025: Developed a score-tracking application for our in-house off-season robotics competitions.
- 2024: Volunteered at our in-house FIRST event and presented our robotics program.
- 2025: *Anges Gardiens* volunteer; peer guidance to new students at Collège Regina Assumpta.
- 2025: Volunteer photographer for dance show at Collège Regina Assumpta in intricate lighting.

Extracurriculars/Notable Projects

- **Medical Inference via Vector Embedding** – Personal research project developing an AI system using vector embeddings to model symptom–condition relationships for disease prediction.
- **HackQC 2024** – Developed a DiscordJS chatbot using OpenAI LLMs and LangChain to query Quebec Open Data, providing users with fast, searchable information.
- 2024 & 2025 ***Secondaire en spectacle*** Performed piano-comedy skits with a friend, 400+ audience. Local winner.

Programming Skills

- **General:** JavaScript/TypeScript, Python, MATLAB
- **Robotics:** Java, Python
- **App development:** Kotlin Multiplatform, SwiftUI
- **AI/ML:** PyTorch, OpenCV, computer vision
- **Web development:** React, HTML, SCSS, OAuth

Other Skills

- Hands-on experience with tools, electronics, and machines
- Basic CAD (Onshape, Autodesk Inventor)
- Photography and Adobe Lightroom
- Proficient with Office suite

Certifications

- Good Clinical Practice (CITI), Éthique de la recherche (MSSS), Responsible Conduct of Research
- **Canadian Red Cross:** Standard First Aid Level & CPR/AED Level C, Psychological First Aid